الأسئلة المقالية — 12 عام · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (36 سؤالاً)

LO21 حل المعادلات التربيعية بإكمال المربع · Solve quadratics by completing the square

21.1 LO21 · Ex2

Complete every part of the question shown in the image. Write your working in the space below.

Ex2	.Solve by completing the square حُلّ المعادلة بإكمال المربع.

$$x^2 - 6x + 12 = 19$$

a. Complete all parts of the question.

الأستاذ كامل بشارة 0502837825

21.2 LO21 · 3

Complete every part of the question shown in the image. Write your working in the space below.

3	.Solve by completing the square حُلّ المعادلة بإكمال المربع.

$$x^2 - 12x + 3 = 8$$

a. Complete all parts of the question.

21.3 LO21 · Ex3

Complete every part of the question shown in the image. Write your working in the space below.

Ex 3	.Solve by completing the square	حُلّ المعادلة بإكمال المربع.

$$-2x^2 + 8x - 18 = 0$$

a. Complete all parts of the question.

21.4 LO21 · 3

Complete every part of the question shown in the image. Write your working in the space below.

3	.Solve by completing the square	حُلّ المعادلة بإكمال المربع.

$$3x^2 - 9x - 3 = 21$$

a. Complete all parts of the question.

21.5 LO21 · Ex5

Complete every part of the question shown in the image. Write your working in the space below.

Ex5	.Solve by completing the square حُلّ المعادلة بإكمال المربع.

$$x^2 + 18x - 4 = 0$$

a. Complete all parts of the question.

21.6 LO21 · Ex7

Complete every part of the question shown in the image. Write your working in the space below.

Ex7	.Solve by completing the square	بإكمال المربع.	حُلّ المعادلة

$$4x^2 - 12x - 27 = 0$$

a. Complete all parts of the question.

21.7 LO21 · english-Q2

Solve by completing the square. Show all steps.

$$x^2 - 18x + 81 = 49$$

a. Complete the square

b. Solve and check

21.8 LO21 · english-Q4

Solve by completing the square. Show all steps.

$$4x^2 - 28x + 49 = 64$$

a. Complete the square

b. Solve and check

21.9 LO21 · english-Q5

Solve by completing the square. Show all steps.

$$25x^2 + 20x + 4 = 75$$

a. Complete the square

b. Solve and check

21.10 LO21 · english-Q8

Solve by completing the square. Show all steps.

$$6x^2 - 21x + 9 = 0$$

a. Complete the square

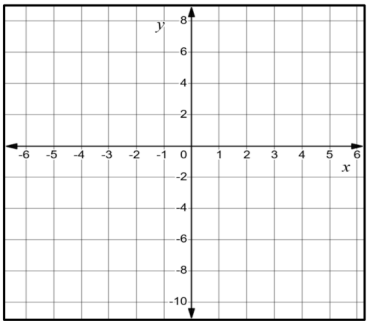
b. Solve and check

22.1 LO22 · Ex2

Complete every part of the question shown in the image. Write your working in the space below.

Ex2 Solve by graphing. حُلّ المتباينة بالتمثيل البياني.

$$x^2 + 2x - 8 < 0$$



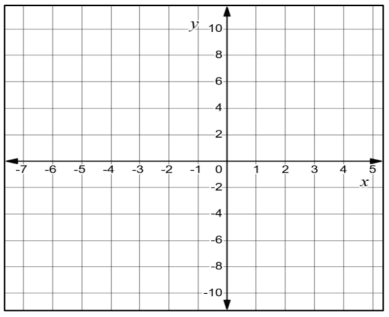
a. Complete all parts of the question.

22.2 LO22 · EX3

Complete every part of the question shown in the image. Write your working in the space below.

EX3 Solve by graphing. حُلّ المتباينة بالتمثيل البياني.

$$2x^2 + 4x - 5 \geq 0$$



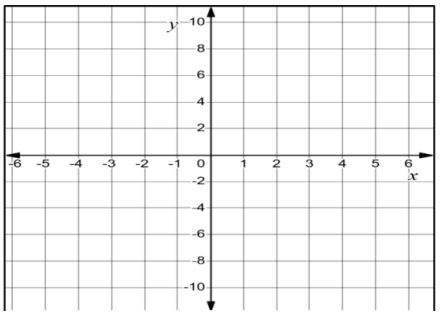
a. Complete all parts of the question.

22.3 LO22 · Ex3

Complete every part of the question shown in the image. Write your working in the space below.

Ex3	Solve by graphing.	حُلّ المتباينة بالتمثيل البياني.

$$x^2 + x - 6 < 0$$



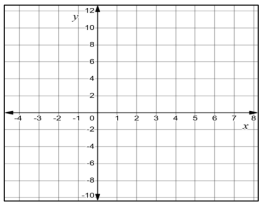
a. Complete all parts of the question.

22.4 LO22 · Ex4

Complete every part of the question shown in the image. Write your working in the space below.

Ex4 Solve by graphing. حل المتباينة بالتمثيل البياني.

$$x^2 - 3x - 4 \geq 0$$



a. Complete all parts of the question.

22.5 LO22 · english-Q11

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$x^2 - 2x - 24 \leq 0$$

a. Five-point table

b. Graph

c. Solution set

22.6 LO22 · english-Q13

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$-x^2 + 3x + 10 < 0$$

a. Five-point table

b. Graph

c. Solution set

22.7 LO22 · english-Q14

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$x^2 - 6x + 2 > 0$$

a. Five-point table

b. Graph

c. Solution set

23.1 LO23 · EX3

Complete every part of the question shown in the image. Write your working in the space below.

EX3			حوّل كل تعبير مما يلي لأبسط صورة.
a)	$(4x^2 - 5x + 6) - (2x^2 + 3x - 1)$	b)	$(6x^2 - 7x + 8) + (-4x^2 + 9x - 5)$

a. Complete all parts of the question.

23.2 LO23 · 32/19

Complete every part of the question shown in the image. Write your working in the space below.

32/19	$(x-y)(x^2+2xy+y^2)$	33	$(a+b)(a^3-3ab-b^2)$

a. Complete all parts of the question.

23.3 LO23 · 38

Complete every part of the question shown in the image. Write your working in the space below.

38	$(x-y)(x+y)(2x+y)$	39/22	$(a+b)(2a+3b)(2x-y)$

a. Complete all parts of the question.

23.4 LO23 · 58/36

Complete every part of the question shown in the image. Write your working in the space below.

58/36	$(2x - 2y)^3$	60/38	$(3z - 2)^3$

a. Complete all parts of the question.

23.5 LO23 · 5

Complete every part of the question shown in the image. Write your working in the space below.

5	$(6a^2 + 5a + 10) - (4a^2 + 6a + 12)$	10	$(-2f^2 - 3f - 5) + (-2f^2 - 3f + 8)$

a. Complete all parts of the question.

23.6 LO23 · english-Q18

Simplify. Show expansion and collection of like terms.

$$(a+b)(b^3-3ab-b^2)$$

a. Expand

b. Combine like terms

23.7 LO23 · english-Q22

Simplify. Show expansion and collection of like terms.

$(4n-5)^3$

a. Expand

b. Combine like terms

23.8 LO23 · english-Q24

Simplify. Show expansion and collection of like terms.

$$(6x^2 - 7x + 8) + (-4x^2 + 9x - 5)$$

a. Expand

b. Combine like terms

23.9 LO23 · english-Q25

Simplify. Show expansion and collection of like terms.

$$(-x^2 - 3x + 4) - (x^2 + 2x + 5)$$

a. Expand

b. Combine like terms

24.1 LO24 · 9/21

Complete every part of the question shown in the image. Write your working in the space below.

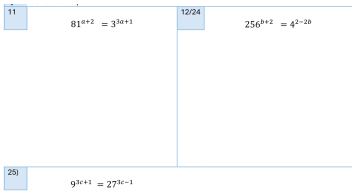
9/21 $8^{4x+2} = 64$

10/22 $5^{x-6} = 125$

a. Complete all parts of the question.

24.2 LO24 · 11

Complete every part of the question shown in the image. Write your working in the space below.



a. Complete all parts of the question.

24.3 LO24 · english-Q30

Solve using a common base. Check whether a solution exists.

$$9^{3c+1} = 27^{2c-1}$$

a. Common base

b. Solve and check

24.4 LO24 · english-Q31

Solve using a common base. Check whether a solution exists.

$$(1/27)^{2d-2} = 81^{d+4}$$

a. Common base

b. Solve and check

24.5 LO24 ·

Solve and show how you express both sides using a common base.

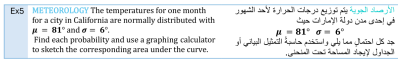
$27^{x-1} = 9^{x+2}$

a. Common base

b. Solve linear equation

25.1 LO25 · Ex5

Complete every part of the question shown in the image. Write your working in the space below.



Ex5 METEOROLOGY The temperatures for one month for a city in California are normally distributed with $\mu = 81°$ and $\sigma = 6°$. Find each probability and use a graphing calculator to sketch the corresponding area under the curve.

الأرصاد الجوية يتم توزيع درجات الحرارة لأحد الشهور في إحدى مدن دولة الإمارات حيث $\mu = 81°$ $\sigma = 6°$. جد كل احتمال مما يلي واستخدم حاسبة التمثيل البياني أو الجداول لإيجاد المساحة تحت المنحنى.

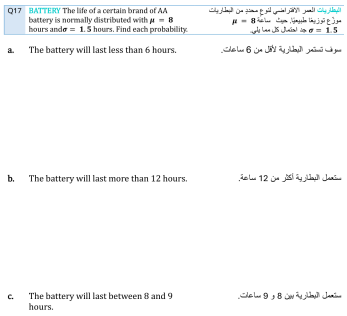
a. $P(70° < X < 90°)$

b. $P(X \geq 95°)$

a. Complete all parts of the question.

25.2 LO25 · Q17

Complete every part of the question shown in the image. Write your working in the space below.



Q17 BATTERY The life of a certain brand of AA battery is normally distributed with $\mu = 8$ hours and $\sigma = 1.5$ hours. Find each probability.

البطاريات العمر الافتراضي لنوع محدد من البطاريات موزع توزيعا طبيعيا، حيث $\mu = 8$ ساعة $\sigma = 1.5$ جد احتمال كل مما يلي

a. The battery will last less than 6 hours.

سوف تستمر البطارية لأقل من 6 ساعات.

b. The battery will last more than 12 hours.

ستعمل البطارية أكثر من 12 ساعة.

c. The battery will last between 8 and 9 hours.

ستعمل البطارية بين 8 و 9 ساعات.

a. Complete all parts of the question.

25.3 LO25 · Ex8

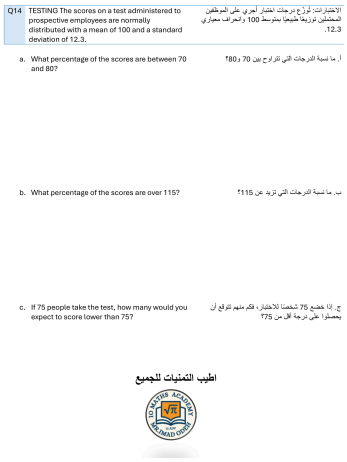
Complete every part of the question shown in the image. Write your working in the space below.



a. Complete all parts of the question.

25.4 LO25 · Q14

Complete every part of the question shown in the image. Write your working in the space below.



Q14	TESTING The scores on a test administered to prospective employees are normally distributed with a mean of 100 and a standard deviation of 12.3.	الاختبارات: تُوزع درجات اختبار أجري على الموظفين المحتملين توزيعا طبيعيا بمتوسط 100 وانحراف معياري 12.3.
a.	What percentage of the scores are between 70 and 80?	أ. ما نسبة الدرجات التي تتراوح بين 70 و80؟
b.	What percentage of the scores are over 115?	ب. ما نسبة الدرجات التي تزيد عن 115؟
c.	If 75 people take the test, how many would you expect to score lower than 75?	ج. إذا خضع 75 شخصا للاختبار، فكم منهم تتوقع أن يحصلوا على درجة أقل من 75؟

اطيب التمنيات للجميع

a. Complete all parts of the question.

25.5 LO25 · english-Q35

Test scores are normally distributed with mean72 and standard deviation5. Use the standard normal table to find P(X<89). Show standardization.

a. Standardize

b. Normal probability